

Miscellaneous/Review

1. (August 2020 Problem 1) On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a non-diagonalizable matrix in $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- (b) Give an example of a nontrivial proper normal subgroup of $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- (c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.
- (d) Give an example of a $\mathbb{Z}$ -module which is flat but not free.
- (e) Give an example of an exact sequence of abelian groups which is not split.

1. Solution

- (a) Let $p = 2$ and consider $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.
- (b) Let $p = 3$. Then $\text{SL}_3(\mathbb{F}_3)$ is a proper normal subgroup of $\text{GL}_3(\mathbb{F}_3)$.
- (c) Let I be the ideal of matrices of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$. Then I is a left ideal but not a right ideal.
- (d) $\mathbb{Q}$
- (e) $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$

2. (January 2021 Problem 1) On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a finite extension of $\mathbb{Q}$ which is not Galois.
- (b) Give an example of a non-trivial, proper, two-sided ideal in $M_2(\mathbb{Z})$.
- (c) Give an example of an injective map $M \rightarrow N$ of $k[x]$ -modules (k a field) and another $k[x]$ -module R such that $M \otimes R \rightarrow N \otimes R$ is not injective.
- (d) Give an example of a commutative ring R and prime ideals p, q, q' in R such that the extension $\bar{q}$ of q to the localization R_p is prime, while the extension $\bar{q}'$ of q' to R_p is not prime.

2. Solution

- (a) $\mathbb{Q}(\sqrt[4]{2})$
- (b) $M_2(2\mathbb{Z})$
- (c) Consider $k[x] \xrightarrow{\cdot x} k[x]$ and $R = k[x]/(x)$. Then $k[x] \otimes_{k[x]} R \cong R \xrightarrow{\cdot x} k[x] \otimes_{k[x]} R \cong R$ is not injective since it is the zero map.

(d) Let $R = \mathbb{Z}$, $p = (2)$, $q = (0)$, and $q' = (3)$.

3. (January 2020 Problem 1) On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

(a) Write down a nonzero element of R which is a zero-divisor.

(b) Write down an element in the center of R which is not in $\mathbb{Z}$.

(c) Write down an R -module which is not a free module.

3. Solution

(a) Let e be the identity in S_3 and let $\sigma = (12)$. Then $e + \sigma$ is a nonzero element of R that is a zero-divisor since $(e + \sigma)(e - \sigma) = 0$.

(b) Let $\tau = (123)$, and consider the element $x = e + \tau + \tau^2$. The element x clearly commutes with τ , so it is enough to check that x commutes with σ . We have

$$\begin{aligned}\sigma x &= \sigma + \sigma\tau + \sigma\tau^2 \\ &= \sigma + \tau^{-1}\sigma + \tau^{-2}\sigma \\ &= \sigma + \tau^2\sigma + \tau\sigma \\ &= x\sigma.\end{aligned}$$

(c) Consider $\mathbb{Z}^3$ as an R -module, where S_3 acts by permuting the entries. So for example, $(2\sigma + \tau)(a, b, c) = (2b + c, 3a, 2c + b)$. For any free R -module M of finite rank we have $\text{rank}_{\mathbb{Z}\text{-mod}}(M) = 6 \text{ rank}_{R\text{-mod}}(M)$, since R has rank 6 as a free $\mathbb{Z}$ -module. However, $\mathbb{Z}^3$ has rank 3 as a $\mathbb{Z}$ -module, which is not divisible by 6, so it cannot be a free R -module.

4. (January 2017 Problem 3) Let R be a commutative ring with unity. Show that a polynomial

$$f(t) = c_n t^n + c_{n-1} t^{n-1} + \cdots + c_0 \in R[t]$$

is nilpotent if and only if all of its coefficients $c_0, \dots, c_n \in R$ are nilpotent.

4. Solution We begin by showing that a sum of two nilpotent elements is nilpotent. Suppose a, b are nilpotent, with least exponents n, m respectively. Then $(a+b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^k b^{n+m-k}$. For each term of this sum, either $k \geq n$, in which case $a^k = 0$, or $k < n$ and $n+m-k > m$, in which case $b^{n+m-k} = 0$. Thus, $(a+b)^{n+m} = 0$, so $a+b$ is nilpotent.

The above immediately shows that if all the coefficients are nilpotent, then in fact $f(t)$ is nilpotent.

Now, suppose $f(t)$ is nilpotent, so that $f^k = 0$ for some $k > 0$. Then looking at the highest degree coefficient, we see that $c_n^k = 0$, so c_n is nilpotent. Thus, also $c_n t^n$ is nilpotent, and so $f(t) - c_n t^n$ is a sum of two nilpotent elements, hence is nilpotent. Repeating the argument now gives that all the coefficients are nilpotent.

5. (January 2021 Problem 4) Let $F/\mathbb{Q}$ be a quadratic extension, and let f and g be two maps of $\mathbb{Q}$ -algebras from F to $M_2(\mathbb{Q})$. (Edit: in this problem we take algebra homomorphisms to be *unital*, meaning that $f(1) = g(1) = I$.)

- (a) Prove that there exists a matrix $X \in \text{GL}_2(\mathbb{Q})$ such that for all $\alpha \in F$, $Xf(\alpha)X^{-1} = g(\alpha)$.
- (b) Prove that there does not exist a choice of f whose image is contained in the set of upper triangular matrices.

5. Solution

- (a) Let $f = \mathbb{Q}[\zeta]$ where ζ is a root of an irreducible quadratic polynomial over $\mathbb{Q}$. Let ζ' be the polynomial's other root, and let $h(x)$ be the polynomial in question.

Let $Z = f(\zeta)$. Then $h(Z)$ must be zero. Thus the minimal polynomial of Z divides $h(Z)$. Therefore ζ and ζ' are the only possible eigenvalues of Z . Since the sum of the eigenvalues is rational, ζ and ζ' are both eigenvalues of Z . Therefore, Z is conjugate to the matrix

$$Z_b = \begin{pmatrix} \zeta & 0 \\ 0 & \zeta' \end{pmatrix}$$

This, any homomorphism f takes k to kI for any rational number k and ζ to XZ_bX^{-1} for some X . This proves part (a).

- (b) Suppose there were. Then $f(\zeta)$ would be some upper triangular matrix with all rational entries whose eigenvalues are ζ and ζ' . However, the eigenvalues of a 2×2 upper triangular matrix are the entries of the diagonal, which must be rational, which is a contradiction.

6. (January 2018 Problem 3) Consider the field with 11 elements $F = \mathbb{F}_{11}$. Let G denote the cyclic group of order 11 (written multiplicatively), with generator $r \in G$. Denote by FG the group algebra of G (also sometimes denoted $F[G]$). We consider r as an element of FG , and let $T: FG \rightarrow FG$ be the F -linear map given by $T(x) = rx$ for all $x \in FG$. Find the Jordan canonical form of T .

6. Solution We'll start by determining the minimal and characteristic polynomials of T . On the one hand, we know that $T^{11} = Id$ so the minimal polynomial $f(y)$ divides $y^{11} - 1$. On the other hand, if the degree of f is less than 11, then $f(T): FG \rightarrow FG$ takes 1 to $f(r) \neq 0$ so $f(T) \neq 0$, so this tells us that $y^{11} - 1$ is the minimal polynomial and since it has degree 11 it is also the characteristic polynomial of T . In characteristic 11, $y^{11} - 1 = (y - 1)^{11}$ so 1 is the only eigenvalue of T and the size of the largest Jordan block with eigenvalue 1 is 11, so the Jordan canonical form of T consists of a single 11×11 Jordan block with eigenvalue 1.

7. (January 2023 Problem 4) Consider the finite field $\mathbb{F}_p$ where p is a prime number.

- (a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p .)

- (b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)

7. Solution

- (a) If f is an irreducible polynomial of degree 5, then it has 5 distinct roots in $\mathbb{F}_{p^5}$, which is the unique degree 5 extension of $\mathbb{F}_p$. Distinct irreducible polynomials will have distinct roots. There are p^5 total elements of $\mathbb{F}_{p^5}$, but p of those elements lie in the base field $\mathbb{F}_p$. Thus, the number of degree 5 elements of $\mathbb{F}_{p^5}$ is $p^5 - p$, and since each irreducible polynomial corresponds to 5 distinct roots, we have that the number of irreducible polynomials is $\frac{p^5 - p}{5}$.
- (b) Saying that $Q(x) \mid P(R(x))$ is the same as saying $P(R(x)) = 0 \pmod{Q(x)}$. But $\mathbb{F}_p[x]/Q(x) \cong \mathbb{F}_{p^d}$ for $d = \deg Q(x)$, where the isomorphism is given by $x \mapsto \alpha$ for α is a root of $Q(x)$ in $\mathbb{F}_{p^d}$. Then saying $P(R(x)) = 0 \pmod{Q(x)}$ is the same as saying $P(R(\alpha)) = 0$ in $\mathbb{F}_{p^d}$. But then $R(\alpha)$ is simply some element in $\mathbb{F}_{p^d}$, and every element of $\mathbb{F}_{p^d}$ can be written as $R(\alpha)$ for some R , so this is the same as saying P has a root in $\mathbb{F}_{p^d}$, which happens if and only if $\deg P \mid \deg Q$.

8. (January 2020 Problem 2) Let k be a field, and let $R = M_2(k)$, the (noncommutative) ring of 2×2 matrices over k .

- (a) Give an example of a simple left R -module M , and a simple right R -module N .
- (b) Can you find another simple left R -module M' which is not isomorphic to M ? (Either find one, or explain why there is none.)
- (c) Compute $\dim_k N \otimes_R M$.

8. Solution

- (a) Let M be the vector space of 2×1 column vectors, and let N be the vector space of 1×2 row vectors.
- (b) Suppose we have a simple left R -module M' . We will show that M' is isomorphic to M . For any nonzero $v \in M'$, we have that Rv is a nontrivial submodule of M' , hence must be all of M' . Fix a nonzero $v \in M'$, and define a map $M' \rightarrow M$ by $Av \mapsto A \begin{pmatrix} 1 & 0 \end{pmatrix}$, for $A \in R$. (Remember, this is well defined since $Rv = M'$.) This map is clearly onto. Notice now that its kernel is a submodule of M' , hence is either 0 or M' . Since this map is not the zero map, the kernel must be 0, and hence this is an isomorphism between M' and M . (There's actually some cool afterlife to this fact: look up "Morita equivalence".)
- (c) As noted above, any vector in M can be written as $A \begin{pmatrix} 1 & 0 \end{pmatrix}$, thus we can rewrite any simple tensor $u \otimes v$ as $uA \otimes \begin{pmatrix} 1 & 0 \end{pmatrix}$. Thus, every tensor in $N \otimes_R M$ can be written as $u \otimes \begin{pmatrix} 1 & 0 \end{pmatrix}$ for some $u \in N$. It is not hard to check that this gives an isomorphism of vector spaces $N \otimes_R M \cong N$, and thus $\dim_k N \otimes_R M = 2$.

9. (August 2016 Problem 2) Let V be a non-zero finite dimensional vector space over $\mathbb{C}$. Given a linear transformation $A: V \rightarrow V$, show that the following are equivalent.

- (i) There exists a linear transformation $P: V \rightarrow V$ such that $P^2 = I$ and $AP = -PA$.
 - (ii) There exists an invertible linear transformation $P: V \rightarrow V$ such that $AP = -PA$.
 - (iii) There exists a direct sum decomposition $V = V_1 \oplus V_2$ such that $AV_1 \subset V_2$ and $AV_2 \subset V_1$.
- (Hint: Thinking about (iii) $\Rightarrow$ (i) helped me to think about (ii) $\Rightarrow$ (iii).)

9. Solution The implication (i) $\Rightarrow$ (ii) is immediate.

For (ii) $\Rightarrow$ (iii), we first begin by showing that for every eigenvalue λ of AP , $-\lambda$ is also an eigenvalue. Suppose $\lambda \neq 0$ is an eigenvalue of AP , and v is an eigenvector for λ . Then notice that $APAv = -A(AP)v = -\lambda Av$. Furthermore, we have that $Av = -P^{-1}APv = -\lambda P^{-1}v \neq 0$, so Av is an eigenvector for AP of eigenvalue $-\lambda$.

Now, divide the eigenvalues of AP into two sets, S_1 and S_2 , where if a nonzero $\lambda \in S_1$, then $-\lambda \in S_2$, and visa versa. Let V_1 be the union of the λ -generalized eigenspaces for all $\lambda \in S_1$, and V_2 that for those $\lambda \in S_2$. Then our calculation in the previous paragraph shows that $AV_1 \subset V_2$ and $AV_2 \subset V_1$, as desired.

Finally, for (iii) $\Rightarrow$ (i), let P be given by the block diagonal matrix $\begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$. It is easy to check this matrix satisfies the properties we desire.